

CHECKPOINT : 3

BATCH : AWS P02

GROUP: A5

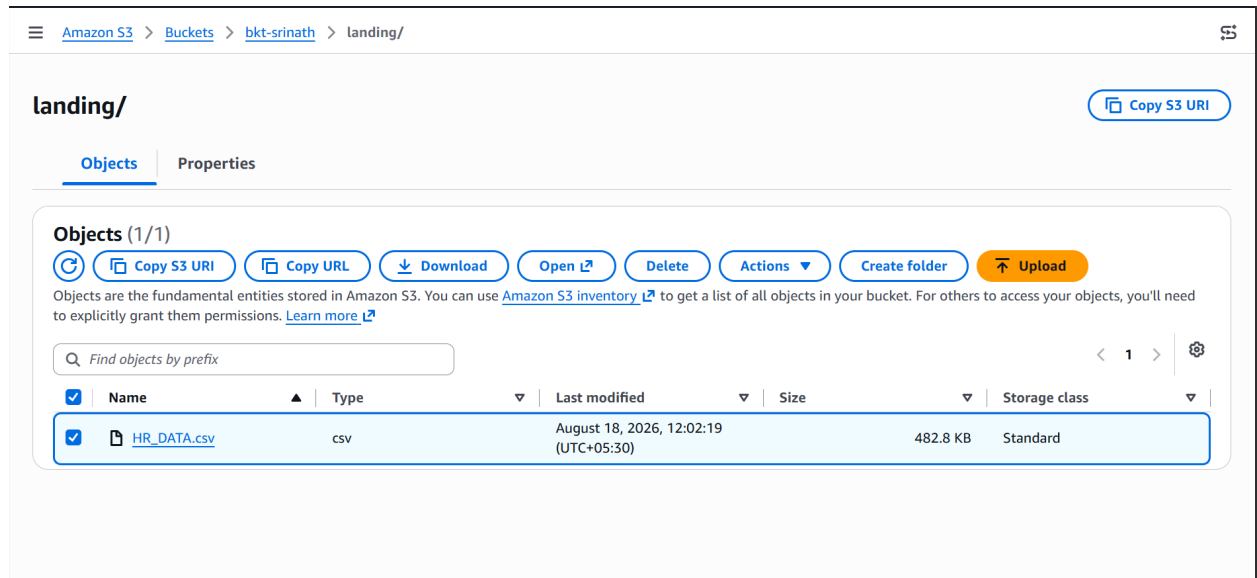
TEAM MEMBERS:

1. Sujay Kumar
 2. Vedant
 3. Ritankar Bhattacharya
 4. Gandikota Charitha
 5. Srinath Neogi
 6. Titas Debnath
-

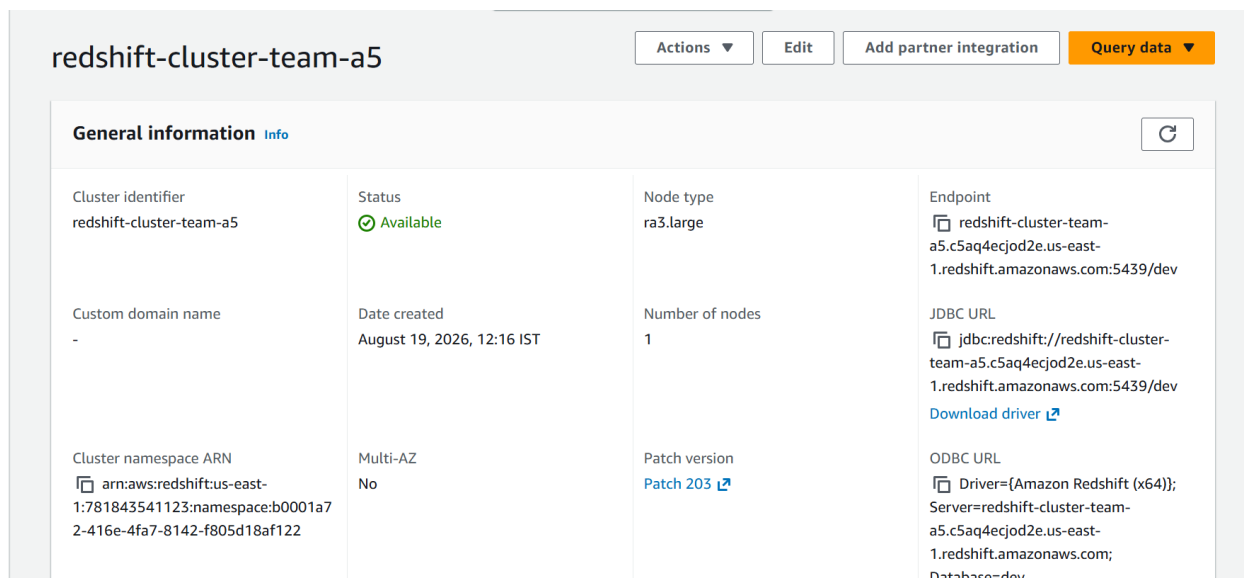
Task 3.1 - Data Analysis + ML Model Training and Deployment on Cloud

1. Redshift to PowerBI Connectivity

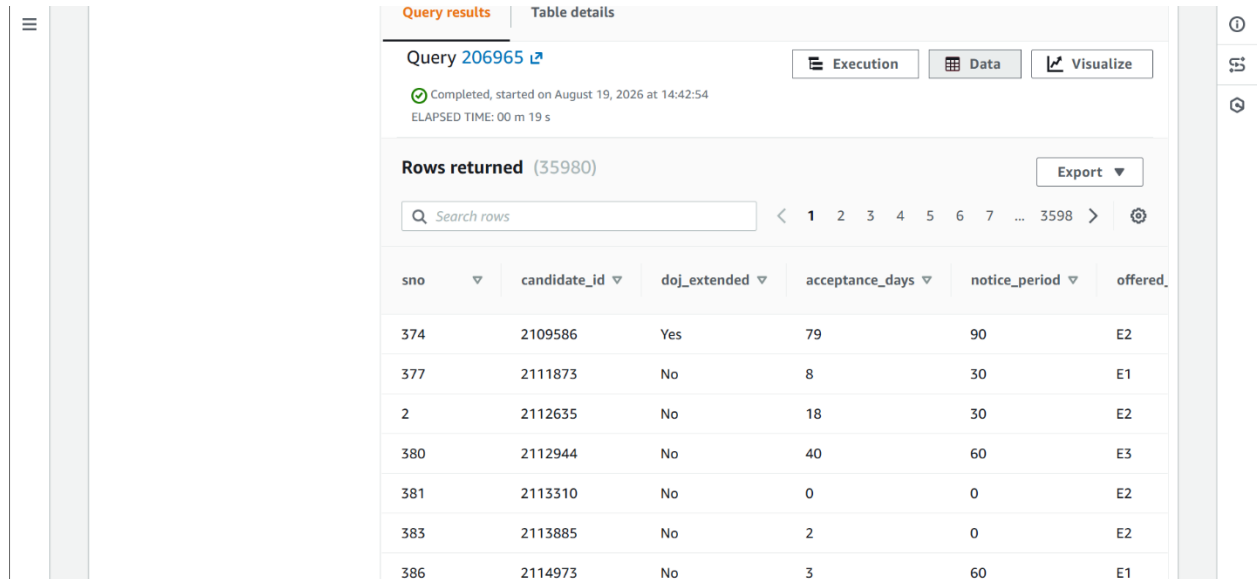
1. Move the Datasets to AWS S3



2. Create Redshift Instance



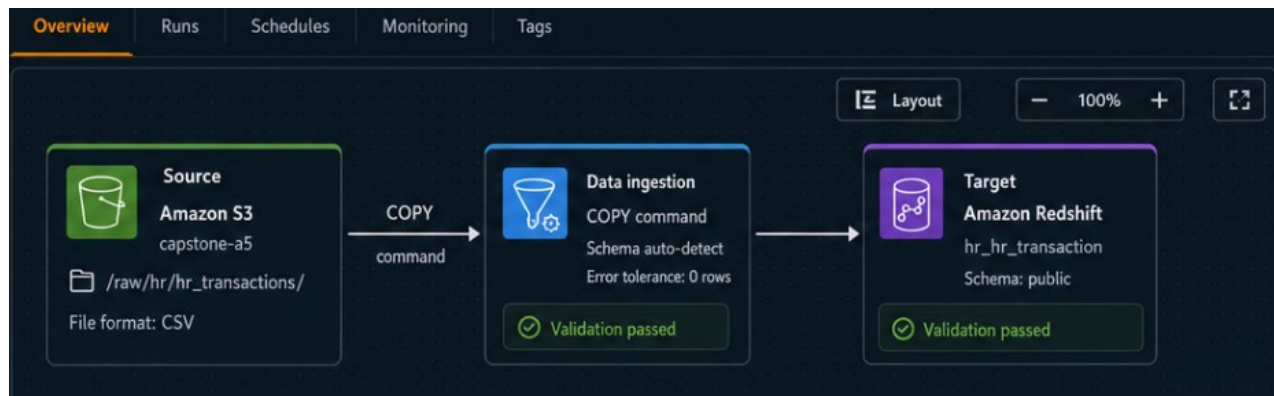
3. Ensure you create required tables in Redshift



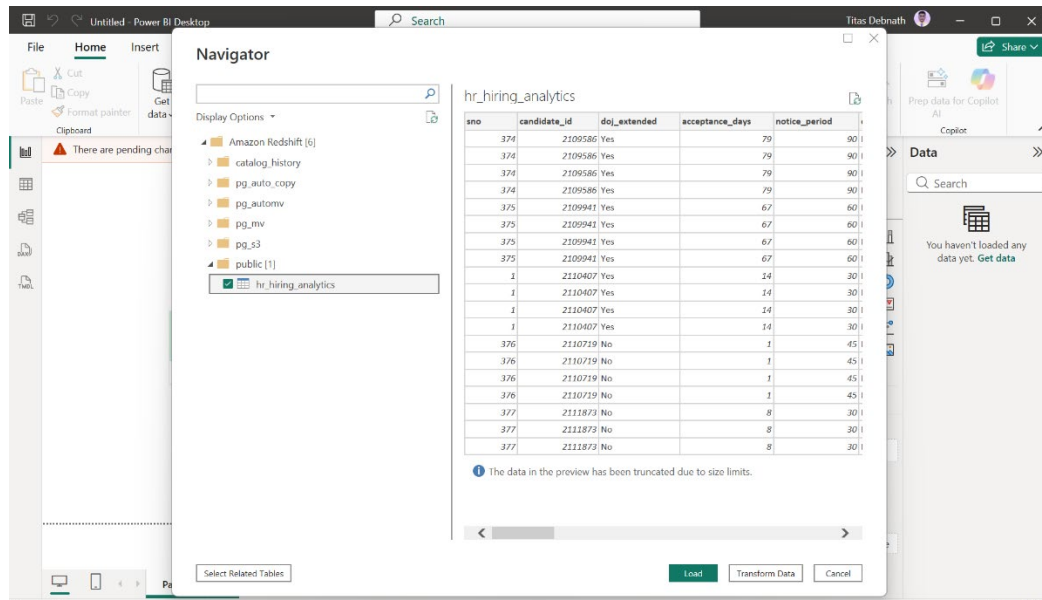
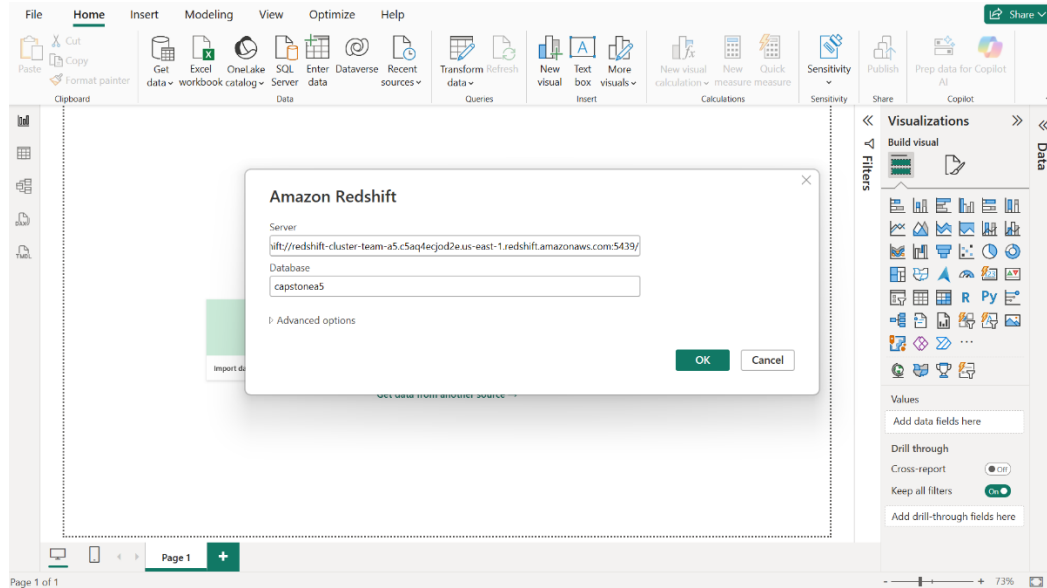
The screenshot displays the Amazon Redshift Query Results interface. At the top, there are tabs for 'Query results' and 'Table details'. The 'Query results' tab is active, showing 'Query 206965' which is 'Completed, started on August 19, 2026 at 14:42:54' with an 'ELAPSED TIME: 00 m 19 s'. Below this, a table shows the first 8 rows of the results. The table has columns: sno, candidate_id, doj_extended, acceptance_days, notice_period, and offered. The 'Rows returned' count is 35,980. There is a search bar and pagination controls.

sno	candidate_id	doj_extended	acceptance_days	notice_period	offered
374	2109586	Yes	79	90	E2
377	2111873	No	8	30	E1
2	2112635	No	18	30	E2
380	2112944	No	40	60	E3
381	2113310	No	0	0	E2
383	2113885	No	2	0	E2
386	2114973	No	3	60	E1

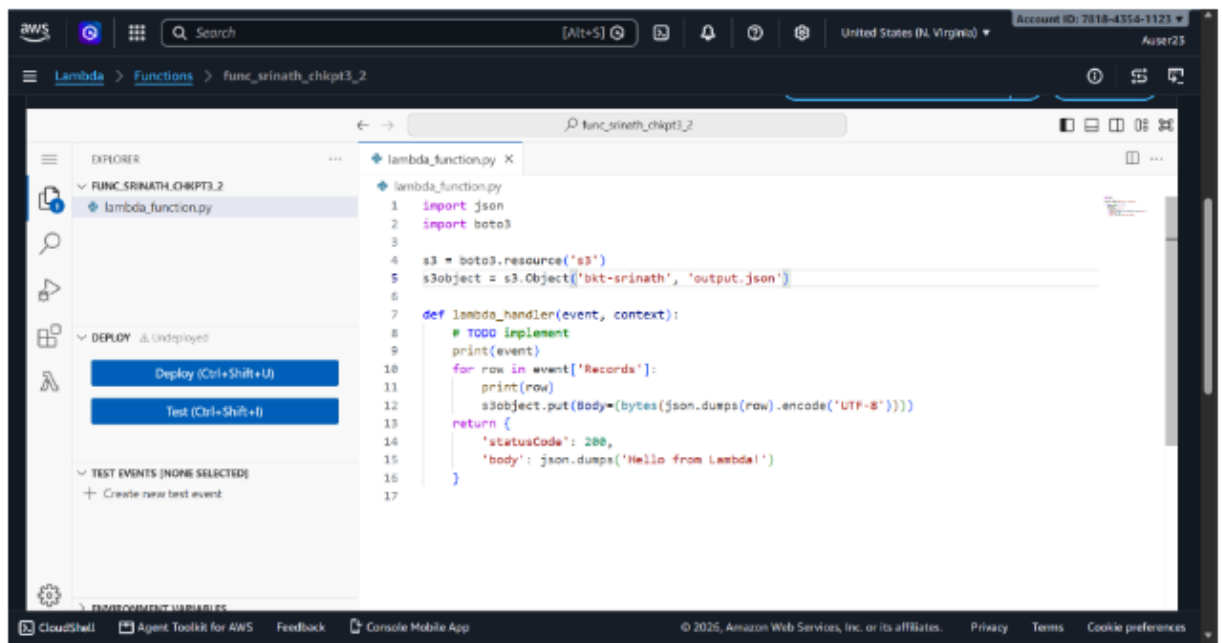
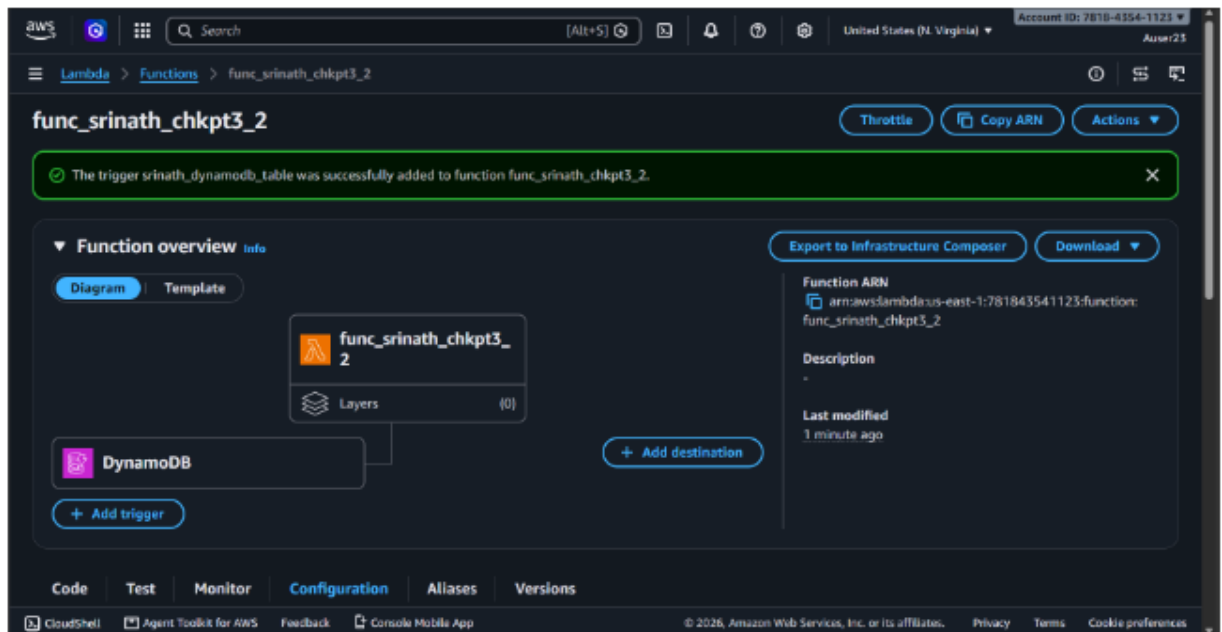
4. Create a data pipeline/copy command to move the data from storage to data warehouse(Redshift). You are allowed to use other copy commands as well to move the data from storage to data warehouse.



5. Connect the Redshift data to PowerBI



2. Dynamodb to S3 bucket configure SNS notifications for any new records added in the Dynamodb



Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

notification-srinath

EditDeletePublish message

Details

Name

notification-srinath

ARN

arn:aws:sns:us-east-1:781843541123:notification-srinath

Display name

-

Type

Standard

Topic owner

781843541123

< Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption >

Subscriptions (1)

EditDeleteRequest confirmationConfirm subscriptionCreate subscription

Search

< 1 >

ID

▲

Endpoint

▼

Status

▼

Protocol

▼

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Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption >

Subscriptions (1)

EditDeleteRequest confirmationConfirm subscriptionCreate subscription

Search

< 1 >

ID

▲

Endpoint

▼

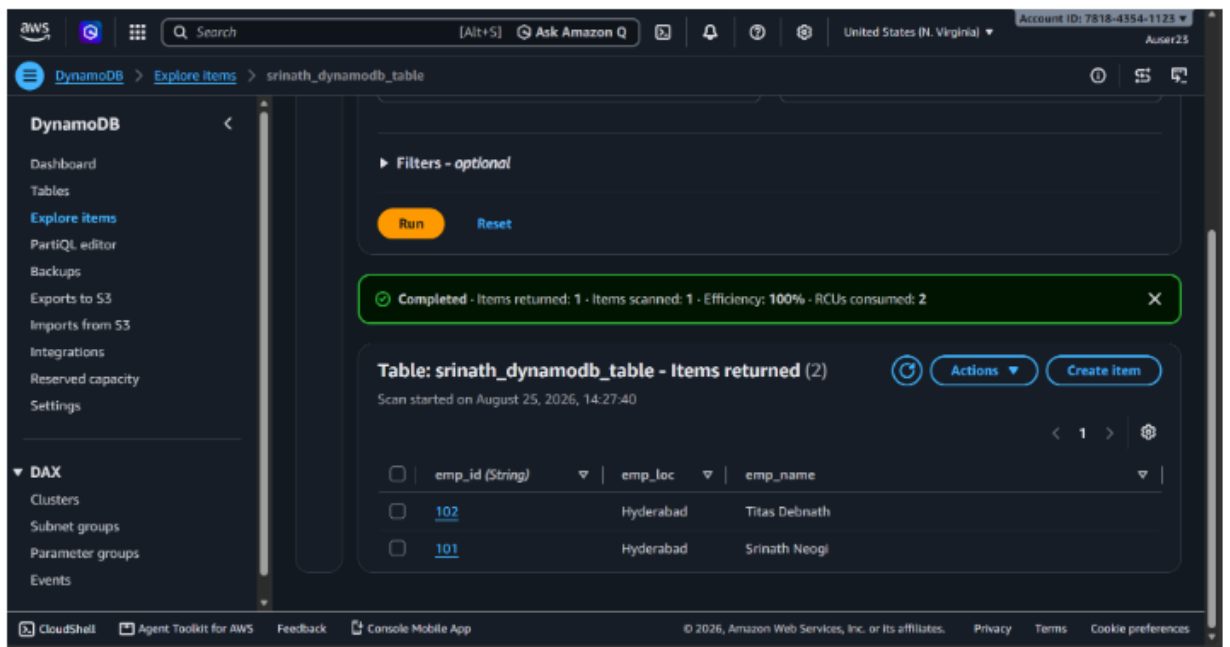
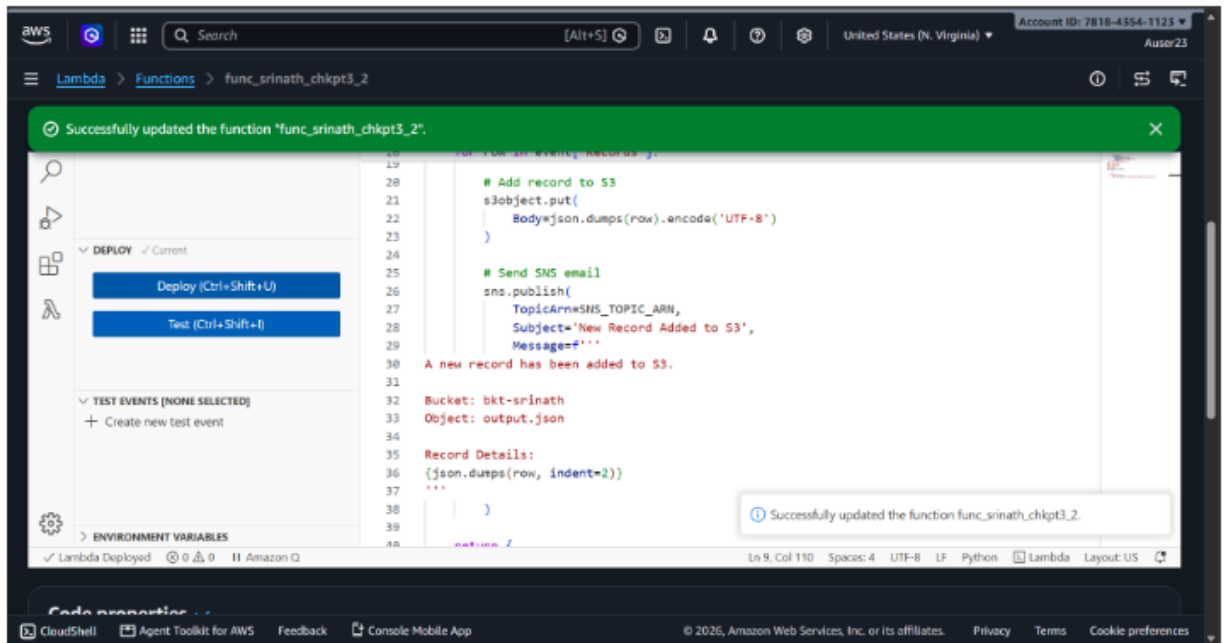
Status

▼

Protocol

▼

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Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Analytics

Log Anomalies

Metrics

Creation time

22 minutes ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Deletion protection

Off

Data protection

-

Sensitive data count

-

Bearer token authentication

Off

Syslog ingestion

Info

Configure

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Log streams (1)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match

Show expired

Info

1

Log stream

Last event time

2026/08/25/[\$LATEST]80d9f700c29343619791233680f45122

2026-08-25 09:36:52 (UTC)

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Amazon S3

Buckets

bkt-srinath

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

1

Name	Type	Last modified	Size	Storage class
output.json	json	August 25, 2026, 15:07:13 (UTC+05:30)	572.0 B	Standard
processed/	Folder	-	-	-

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[EXT] New Record Added to S3



AWS Notifications <no-reply@sns.amazonaws.co

To: Neogi, Srinath



Reply

Reply All

Forward



Tue 8/25/2026 3:19 PM

Links and other functionality have been disabled in this message. To turn on that functionality, move this message to the Inbox.

[You don't often get email from no-reply@sns.amazonaws.com. Learn why this is important at <https://aka.ms/LearnAboutSenderIdentification>]

A new record has been added to S3.

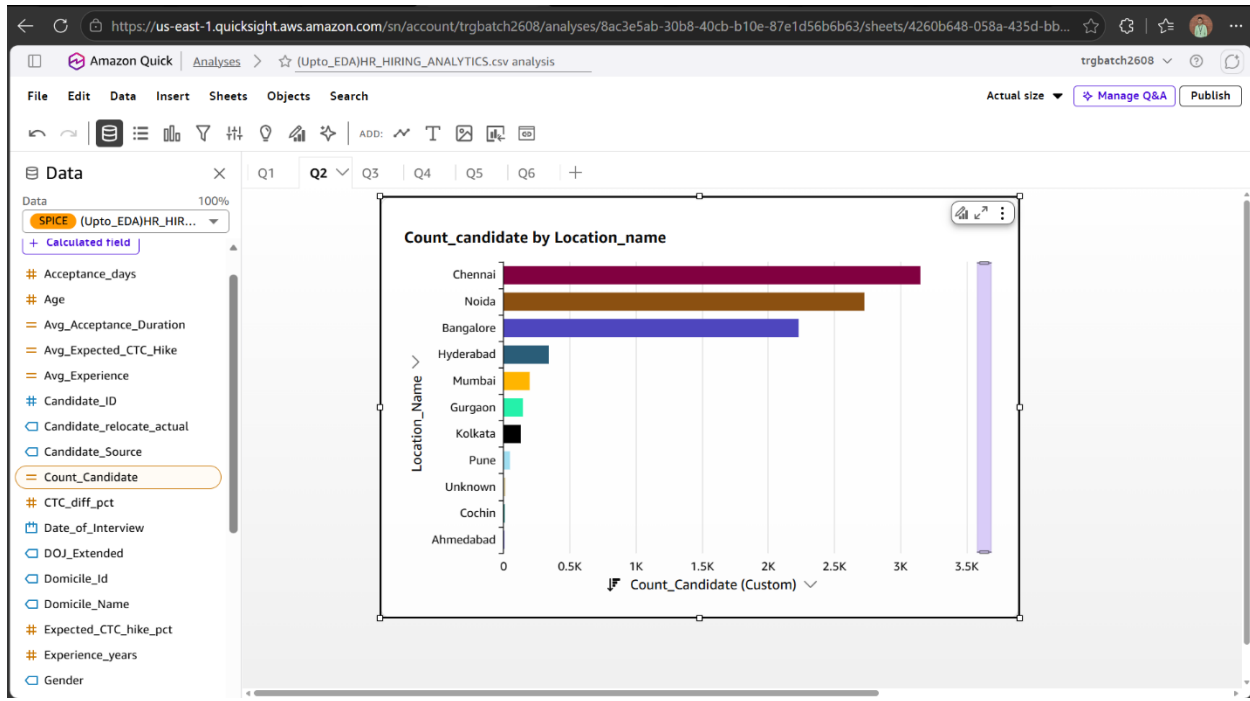
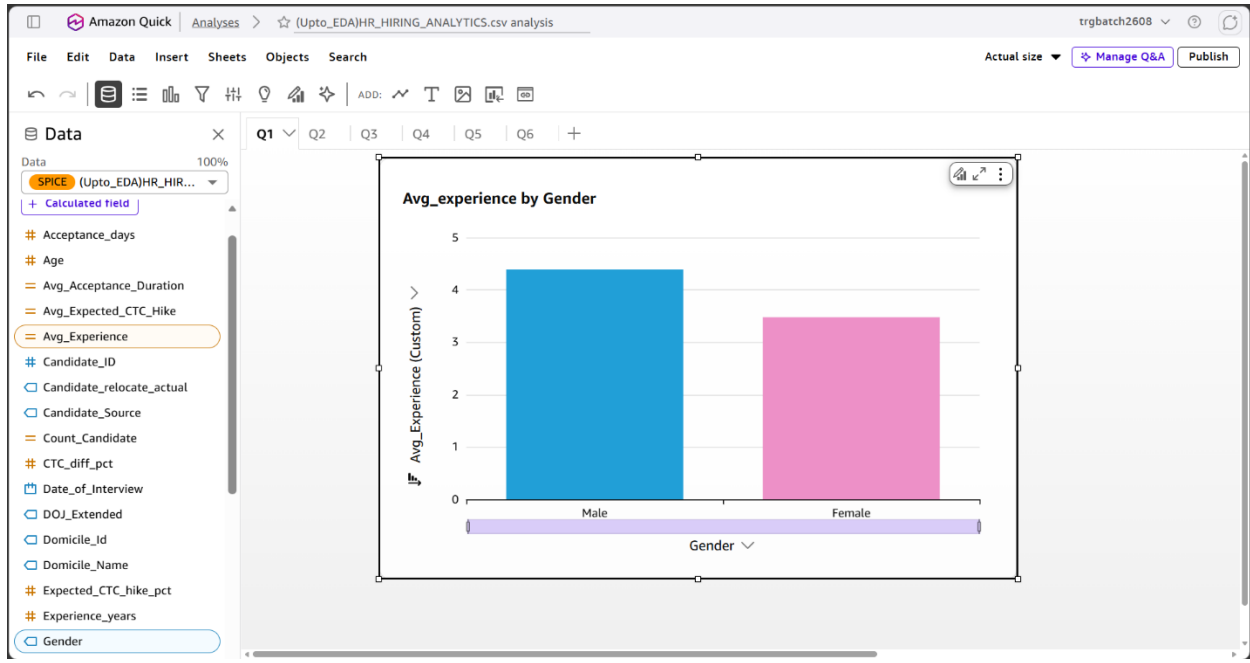
Bucket: bkt-srinath

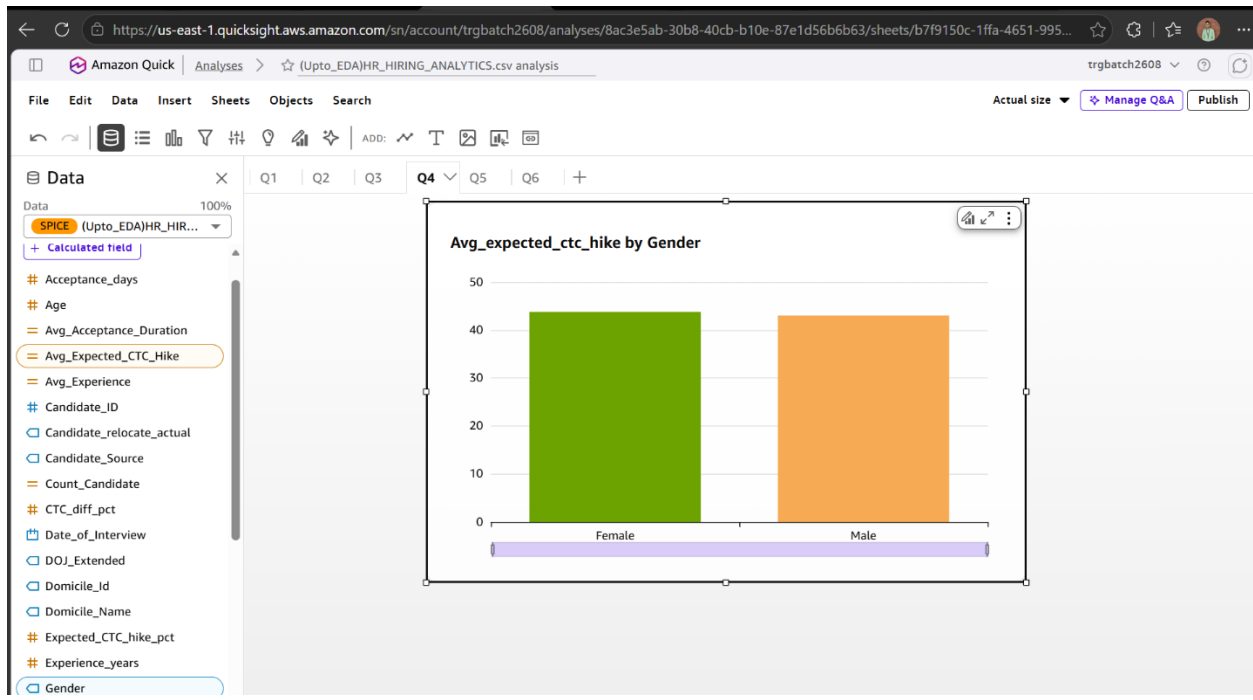
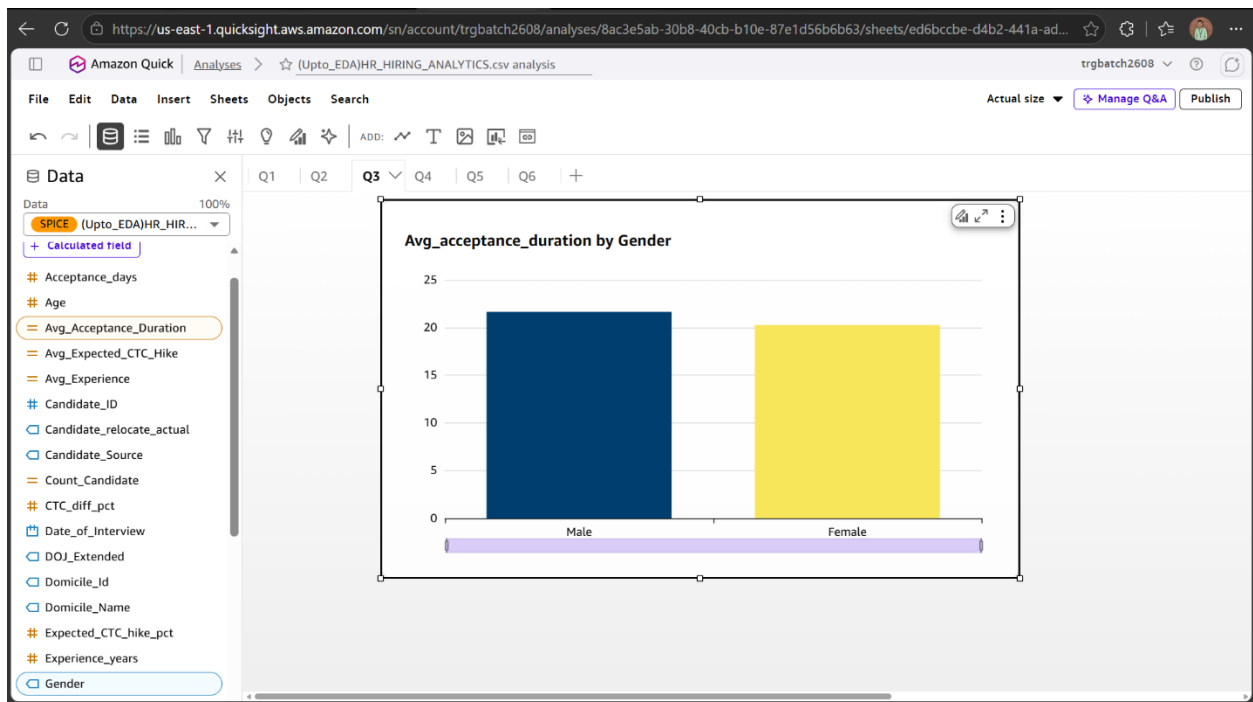
Object: output.json

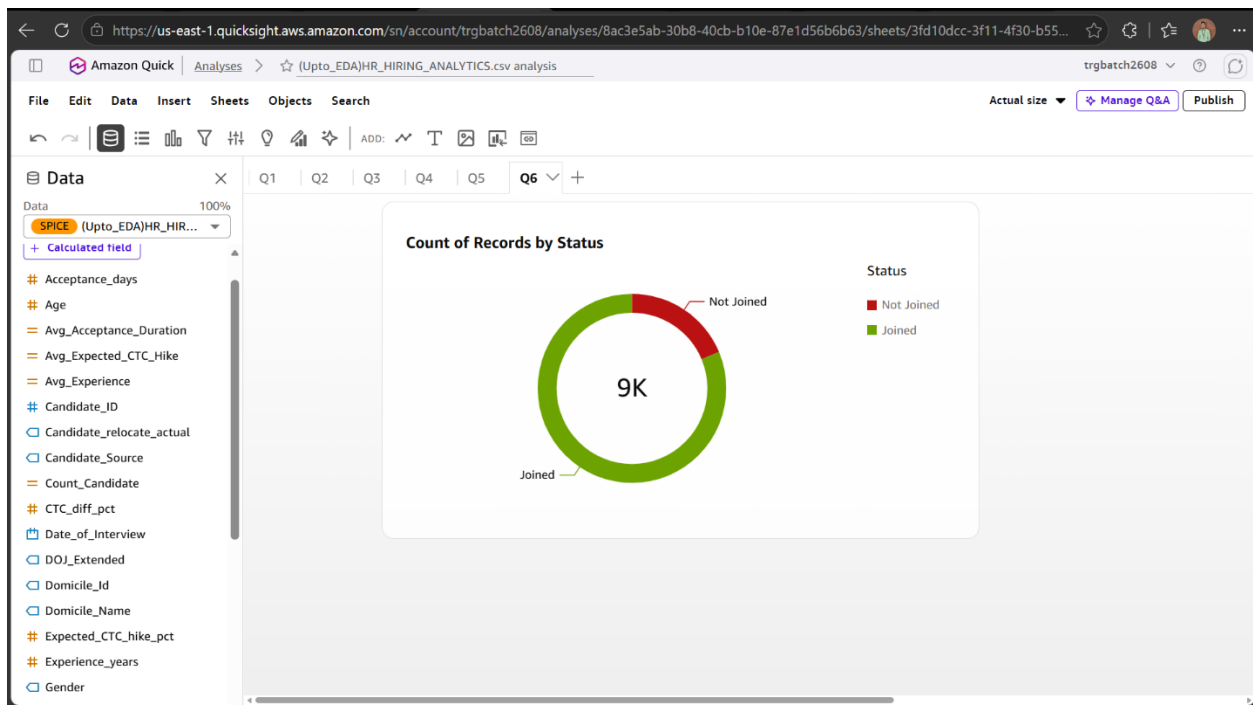
Record Details:

```
{
  "eventID": "eef3e2b9529271904f6993d93ad8a507",
  "eventName": "INSERT",
  "eventVersion": "1.1",
  "eventSource": "aws:dynamodb",
  "awsRegion": "us-east-1",
  "dynamodb": {
    "ApproximateCreationDateTime": 1787651339.0,
    "Keys": {
      "emp_id": {
        "S": "104"
      }
    },
    "NewImage": {
      "emp_loc": {
```

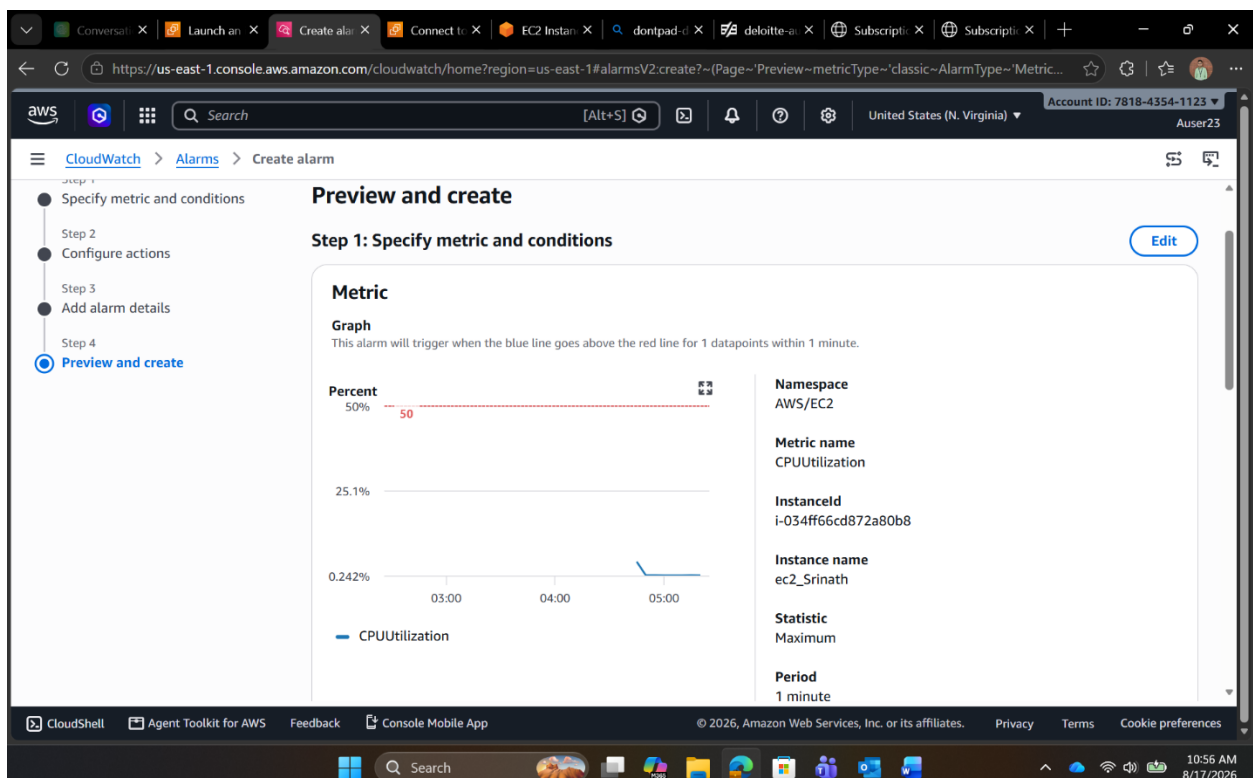
3. Transfer the AWS S3 data to AWS Quicksight perform the same analysis done using powerBI(Any 5 core reports)

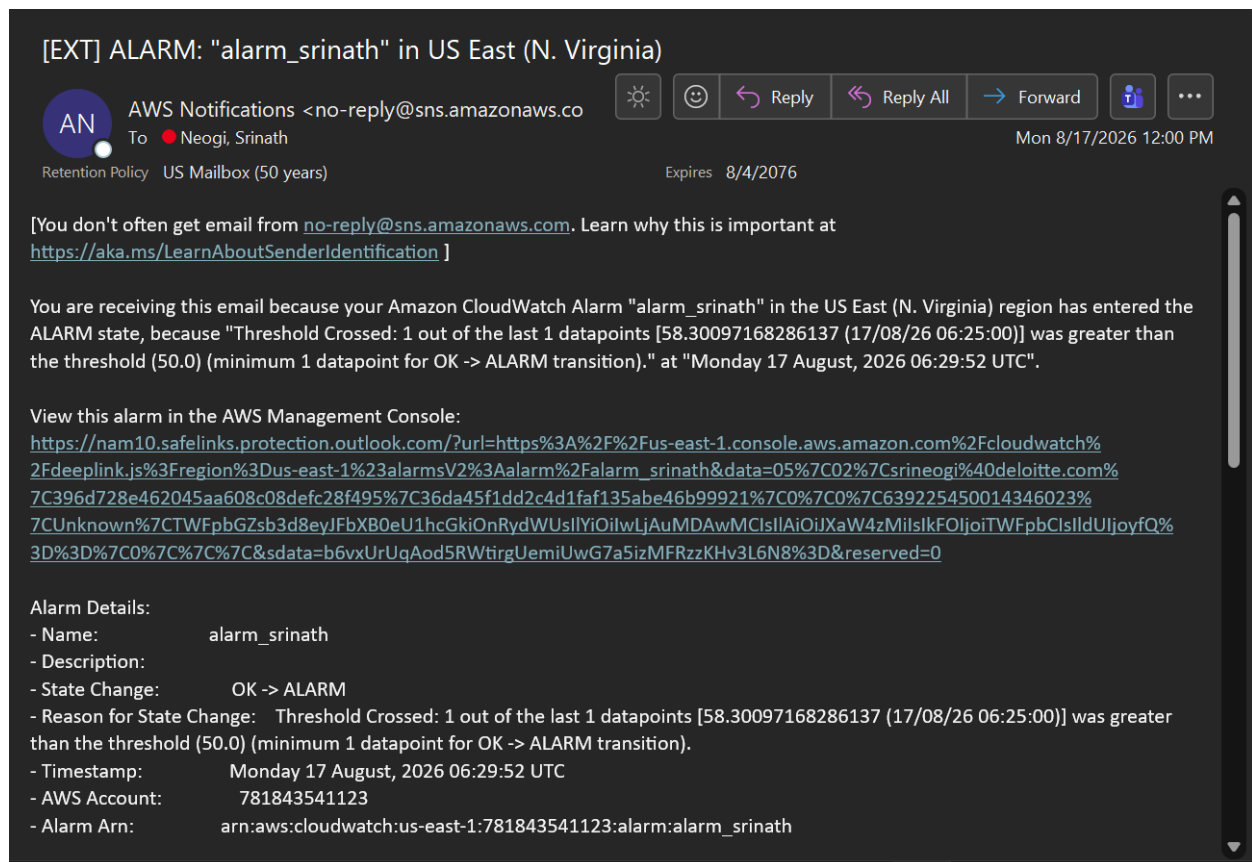




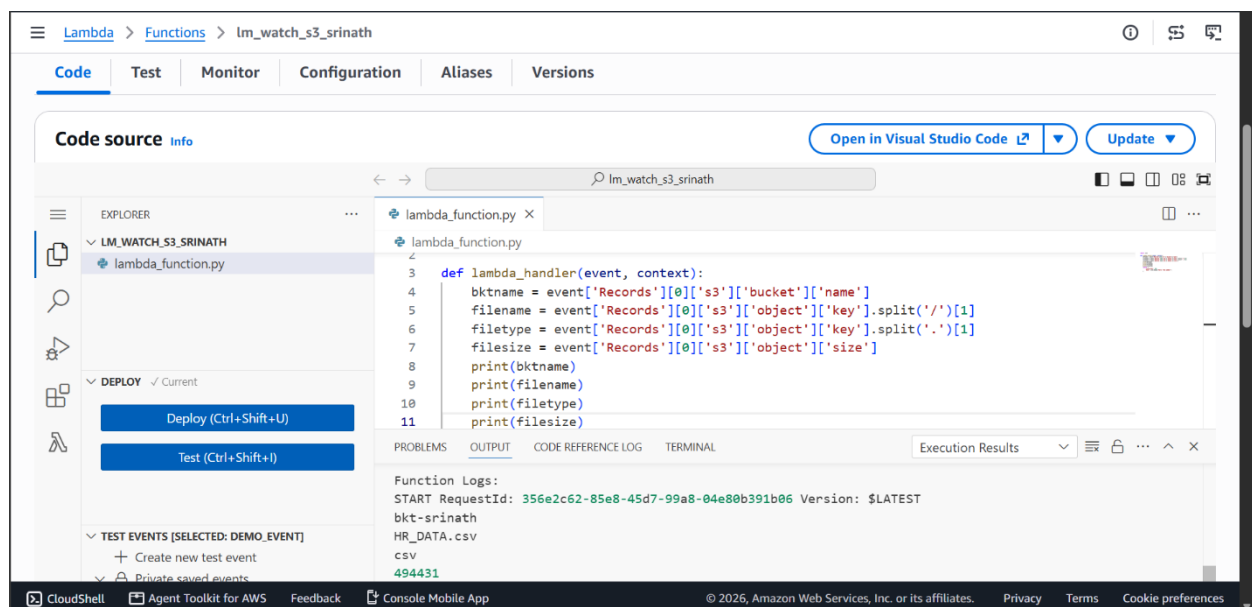


4. 50% of storage exceeds then cloud watch has to trigger the alarm



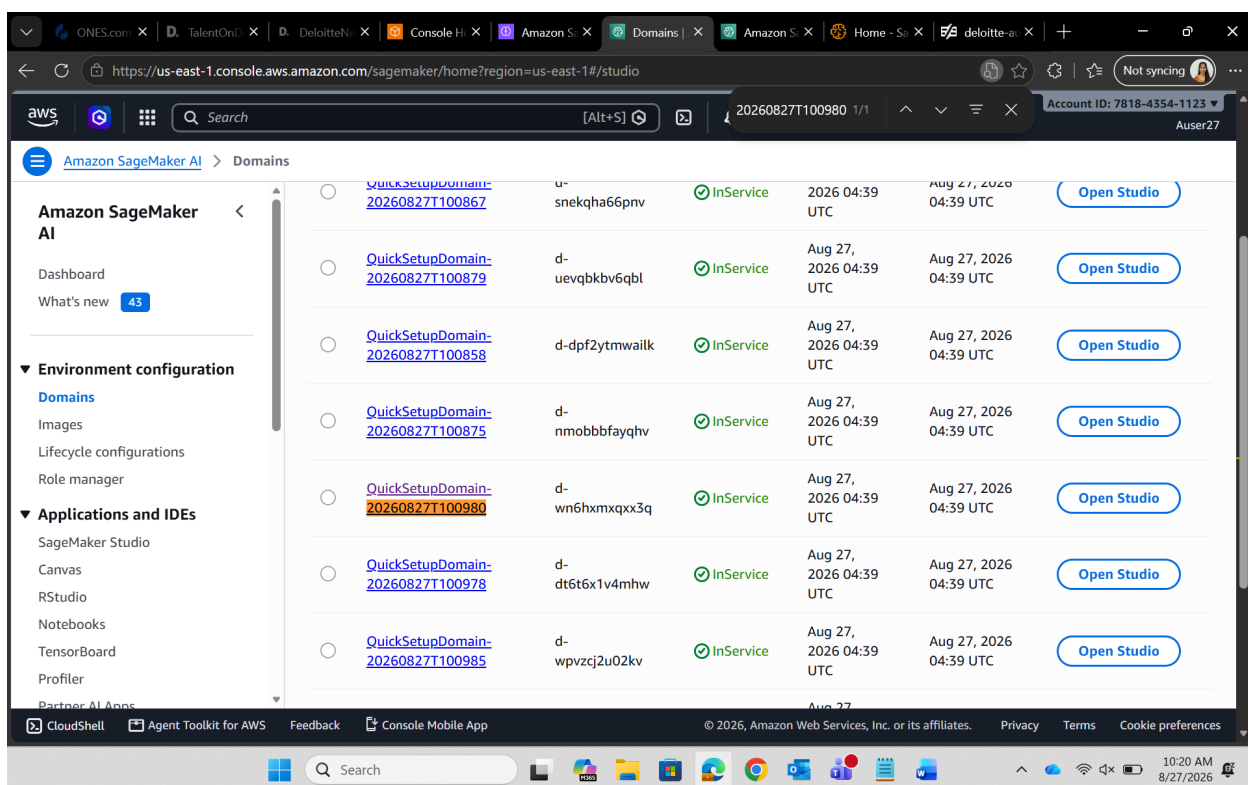


5. Write a Lambda function which logs in cloud trail about S3 file type and size.



```
lambda_function.py X
lambda_function.py
2
3 def lambda_handler(event, context):
4     bktname = event['Records'][0]['s3']['bucket']['name']
5     filename = event['Records'][0]['s3']['object']['key'].split('/')[1]
6     filetype = event['Records'][0]['s3']['object']['key'].split('.')[1]
7     filesize = event['Records'][0]['s3']['object']['size']
8     print(bktname)
9     print(filename)
10    print(filetype)
11    print(filesize)
```

6. Build appropriate ML model/s on the data using AWS Sagemaker , Identify the right metric to evaluate the performance of the model and Deploy on AWS Sagemaker



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https://studio-d-wn6hmxqx3q.studio.us-east-1.sagemaker.aws/home?tab=Overview

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sagemaker-scikit-learn-2025-08-26	Training	Completed	365 days ago	View
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sagemaker-scikit-learn-2025-08-26	Training	Completed	365 days ago	View

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https://us-east-1.console.aws.amazon.com/sagemaker/home?region=us-east-1#/studio

aws | Search | [Alt+S] | 20260827T100980 1/1 | Account ID: 7818-4354-1123 | Auser27

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Name	Id	Status	Created on	Modified on	Actions
QuickSetupDomain-20260827T100867	d-snekqha66pvn	InService	Aug 27, 2026 04:39 UTC	Aug 27, 2026 04:39 UTC	Open Studio
QuickSetupDomain-20260827T100980	d-wn6hmxqx3q	InService	Aug 27, 2026 04:39 UTC	Aug 27, 2026 04:39 UTC	Open Studio
QuickSetupDomain-20260827T100978	d-dt6t6x1v4mhw	InService	Aug 27, 2026 04:39 UTC	Aug 27, 2026 04:39 UTC	Open Studio

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QuickSetupDomain-20260827T100980

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SageMaker Studio > Jupyterlab

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File Edit View Run Kernel Git Tabs Settings Help

default-20260827100980 / quickstart-default-ulnwoj

Launcher

Notebook

- Python 3 (ipykernel)
- Glue PySpark
- Glue Spark
- SparkMagic PySpark
- SparkMagic Spark

Console

- Python 3 (ipykernel)
- Glue PySpark
- Glue Spark
- SparkMagic PySpark
- SparkMagic Spark

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